

JOHN FERRAIOLI

DATA SCIENTIST & APPLICATION DEVELOPER

Philadelphia, PA // New York, NY | ferraiolijohn@gmail.com | (917) 679-9184 | [linkedin.com/in/john-ferraioli/](https://www.linkedin.com/in/john-ferraioli/)

EDUCATION

Bachelor of Science, Data Science – 3.78 GPA

Sep. 2020 - Jun. 2025

Drexel University – Philadelphia, PA

- Minor in Mathematics

PROFESSIONAL EXPERIENCE

Application Engineer Intern, *The Vanguard Group, Inc.*

Sep. 2023 - Mar. 2024

- Built core services for brokerage order management application using Java and Spring Boot, including an order tagging engine and web interface used internally to manage order routing rules.
- Boosted reliability and test coverage by writing 50+ unit tests in JUnit, contributing to a 100% code coverage rate and reducing pre-release bugs.
- Enabled business-side testing by implementing Cucumber BDD tests, allowing non-technical stakeholders to validate application behavior without developer involvement.
- Validated functionality through endpoint testing with Postman and ensured smooth deployments using Bitbucket and Bamboo.
- Improved observability and diagnostics by integrating OpenTelemetry with SplunkCloud and building Honeycomb dashboards, increasing developer efficiency and issue traceability.

Data Management Intern, *Brandywine Global Investment Management, LLC*

Sep. 2022 - Mar. 2023

- Achieved scalable data integration by building Dell Boomi pipelines to manage 100+ Snowflake schemas and tables.
- Improved pipeline efficiency by automating job scheduling and execution using Workload Automation iXp, streamlining end-to-end data flow.
- Ensured data integrity by performing Snowflake SQL quality checks, identifying and resolving issues such as duplicates and null values across key datasets.
- Accelerated data analysis by optimizing complex SQL queries and stored procedures, transforming raw data into curated tables for downstream analytics teams.

Data Analytics Intern, *Imagine360*

Oct. 2021 - Apr. 2022

- Improved claim data quality using SQL Server Management Studio to clean and prep 60,000+ healthcare claims for analysis.
- Streamlined daily operations by developing reusable R scripts to transform raw data into readable, structured datasets.
- Enhanced pricing forecasts by applying a reference-based pricing model to estimate future costs for client healthcare plans.

PROJECTS

BenchQuest, *Drexel University*

(Ongoing)

- Enabled cross-device gameplay by designing and implementing cloud-based AWS DynamoDB database for a fitness app that translates workout progress into game rewards.
- Achieved reliable data storage by building backend logic to persist user data, including profiles, fitness activity, and in-game progress (e.g., avatar levels, gold).
- Improved development clarity and collaboration by creating detailed UML diagrams in Draw.io to outline the app's database architecture.

SKILLS

-
- **Languages & Frameworks:** Python (NumPy, Pandas, Scikit-learn), R (Tidyverse), SQL, Java (Spring Boot).
 - **Data Engineering & Storage:** Snowflake, BigQuery, DynamoDB, PostgreSQL, Microsoft SQL Server Management Studio.
 - **Machine Learning & AI:** Scikit-learn (traditional ML), XGBoost / LightGBM (boosting algorithms).
 - **Cloud:** AWS (S3, EC2, Lambda), Google Cloud (BigQuery, AI Platform), Dell Boomi.
 - **Data Visualization & BI Tools:** Matplotlib / Seaborn, ggplot2.
 - **Version Control & Collaboration:** GitHub/Bitbucket, Jupyter Notebooks, Google Colab, Perforce.